Infinite finitely presented residually finite *p*-groups for every prime

1. Statement and conventions

Throughout, p is a prime and F_p is the field with p elements. All groups are abstract groups. A p-group means a group in which every element has finite p-power order; the group itself need not be finite. A finite presentation means finitely many generators and finitely many relators.

A group G is residually finite if, for every $g \neq 1$, there is a homomorphism from G to a finite group whose value at g is not the identity. Let

$$R = \mathrm{F}_p \oplus I, \quad I = \oplus_{r \geq 1} I_r$$

be a graded unital associative algebra, with $I_r I_s \subseteq I_{r+s}$. Each element is a finite sum of homogeneous terms. Write

$$I_{\geq d} = \oplus_{r \geq d} I_r, \quad R_d = R/I_{\geq d} \quad (d \geq 1).$$

The ideal I is *matrix-nil* if every matrix in $\mathrm{M}_r(I)$ is nilpotent, for every finite r. The nilpotence exponent may depend on the matrix. Steinberg groups, their stable limit, and their matrix maps are defined in Section 2.

Theorem 1.1 (Existence for every prime). *For every prime p there exists an infinite finitely presented residually finite abstract p-group. More precisely, let R(p) be the finite word algebra constructed in Section 16 over F_p. For every $n \geq 12$ the group*

$$G_p = \ker[\mathrm{St}_n(R(p)) \to \mathrm{St}_n(\mathrm{F}_p)]$$

has all four properties. In particular one may take n = 12 for every prime.

Theorem 18.1 proves that this particular algebra is infinite-dimensional, finitely presented, graded, and matrix-nil, with finite-dimensional homogeneous parts. Theorem 22.2 proves stable separation by its degree truncations. Sections 10-14 supply the group-theoretic steps, and Section 4 assembles them. The exponent is allowed to depend on the group element.

2. Steinberg groups and natural maps

For a unital associative ring B and $n \geq 3$, define $\mathrm{St}_n(B)$ by generators $\mathrm{x}_{ij}(b)$, for $1 \leq i \neq j \leq n$ and $b \in B$, and the relations

$$\mathrm{x}_{ij}(a)\mathrm{x}_{ij}(b) = \mathrm{x}_{ij}(a + b),$$

$$[\mathrm{x}_{ij}(a), \mathrm{x}_{kl}(b)] = 1 \quad \text{if } j \neq k \text{ and } i \neq l,$$

$$[\mathrm{x}_{ij}(a), \mathrm{x}_{jk}(b)] = \mathrm{x}_{ik}(ab) \quad \text{if } i, j, k \text{ are distinct.}$$

Here $[x, y] = xyx^{-1}y^{-1}$. Root addition gives $\mathrm{x}_{ij}(0) = 1$ and $\mathrm{x}_{ij}(a)^{-1} = \mathrm{x}_{ij}(-a)$.

Let E_{ij} denote the matrix unit and let $\mathrm{E}_n(B)$ be the subgroup of $\mathrm{GL}_n(B)$ generated by the elementary matrices $1_n + b\mathrm{E}_{ij}$. Multiplication of matrix units verifies the displayed relations, so there is a surjective homomorphism

$$\pi_n : \mathrm{St}_n(B) \to \mathrm{E}_n(B), \quad \mathrm{x}_{ij}(b) \mapsto 1_n + b\mathrm{E}_{ij}.$$

A unital ring homomorphism acts on root parameters and thus induces a homomorphism on Steinberg groups. Adjoining one coordinate also gives a homomorphism $\mathrm{St}_n(B) \to \mathrm{St}_{n+1}(B)$. No injectivity of these maps is built into their definition.

Define $\mathrm{St}(B)$ as the direct limit of these groups. The elementary matrix groups embed by adjoining identity blocks; their union is denoted $\mathrm{E}(B)$. The maps π_n induce

$$\pi : \mathrm{St}(B) \to \mathrm{E}(B), \quad J(B) = \ker \pi.$$

We shall use naturality: applying a ring map and then stabilizing has the same effect as first stabilizing and then applying the ring map. This holds on every root generator and therefore on every group element.

Lemma 2.1. *For every $n \geq 3$, $St_n(B)$ is perfect. Also, for fixed $i \neq j$, distinct root parameters define distinct group elements.*

Proof. Choose k different from i, j. The consecutive-root relation gives

$$\mathrm{x}_{ij}(a) = [\mathrm{x}_{ik}(a), \mathrm{x}_{kj}(1)].$$

Every generator is therefore in the commutator subgroup. If $\mathrm{x}_{ij}(a) = \mathrm{x}_{ij}(b)$, applying π_n and comparing the (i, j) entries gives $a = b$. □

For the graded algebra in Section 1, projection to degree zero is a unital ring homomorphism $R \to \mathrm{F}_p$. It supplies the reduction maps in Theorem 1.1. Moreover $R_1 = \mathrm{F}_p$.

3. Finite-index rewriting

Lemma 3.1. *A subgroup of finite index in a finitely presented group is finitely presented.*

Proof. Write $\Gamma = F/N$, where F is the free group on a finite set X and N is the normal closure of a finite set of relators $\mathscr{R}$. Let $H \leq \Gamma$ have index q, and let F_0 be its inverse image in F. Then $[F : F_0] = q$, $N \subseteq F_0$, and $H \cong F_0/N$.

Make the finite coset graph whose vertices are the right cosets $F_0 f$. For each $x \in X$, put a directed edge labeled x from $F_0 f$ to $F_0 fx$. Traversing an edge backwards reads x^{-1}. The graph is connected because X generates F. Choose a spanning tree rooted at the identity coset.

For each vertex choose the word read along its tree path from the root. These words form a finite right-coset transversal T, containing 1. For a word f, denote the representative of its coset by $\mathrm{t}(f)$. For $t \in T$ and $x \in X$, put

$$y(t, x) = tx\ \mathrm{t}(tx)^{-1} \in F_0.$$

An edge in the spanning tree gives the trivial element. Each edge outside the tree gives one generator, with its reverse giving the inverse generator. Thus there are finitely many nontrivial generators, at most $q|X|$.

To see directly that these freely generate F_0, read a word from the root and insert the tree path to its current vertex and its inverse between consecutive letters. If the original word ends at the root, the inserted tree paths telescope and express the word as a product of the displayed generators and their inverses. Conversely, expand a product of these generators into its edge paths. A reduced nonempty word in the edges outside the tree cannot disappear by cancelling consecutive inverse edges: after all tree excursions are removed, that same reduced edge word remains. It therefore cannot label the identity element of the free group F. This proves freeness.

As a subgroup of F_0, the group N is the normal closure of the finite set

$$\{\, trt^{-1} : t \in T, r \in \mathscr{R} \,\}.$$

Indeed, these elements lie in N. Conversely, each defining conjugate frf^{-1} for its normal closure in F can be written with $f = ht$, where $h \in F_0$ and $t \in T$. It then equals $h(trt^{-1})h^{-1}$. Rewrite each of the finitely many words trt^{-1} in the free generators above. These are finitely many relators for $F_0/N \cong H$. □

4. Proof of the existence theorem

Proof of Theorem 1.1. Fix p and $n \geq 12$. Take $R = R(p)$ from Theorem 18.1, put $\Gamma = \mathrm{St}_n(R)$, and let G_p be the kernel of degree-zero reduction. We prove all four properties for this same subgroup.

Finite presentation

Corollary 13.3 with $d = 1$ makes $\mathrm{St}_n(\mathrm{F}_p)$ finite. The index of G_p in Γ equals the size of the image of the reduction map and is therefore finite. Theorem 10.1 and Lemma 3.1 give a finite presentation of G_p.

Infinitude

Each $\mathrm{x}_{12}(a)$, with $a \in I$, reduces to $\mathrm{x}_{12}(0) = 1$, so belongs to G_p. Distinct values of a give distinct group elements by Lemma 2.1. Since R is infinite-dimensional over the finite field F_p, its positive-degree ideal I is infinite. Hence G_p is infinite.

Residual finiteness

Let $g \in G_p$ be nonidentity. Theorem 12.1 makes its image in $\mathrm{St}(R)$ nonidentity. Theorem 22.2 detects it in $\mathrm{St}(R_d)$ for some d. Naturality gives the same stable image if one first maps g to $\mathrm{St}_n(R_d)$. That fixed-rank image cannot be the identity. It lies in a finite group by Corollary 13.3, giving the required finite quotient image.

Elementwise p-power torsion

For $g \in G_p$, its matrix image has the form $1_n + A$, with $A \in \mathrm{M}_n(I)$. Choose N such that $A^N = 0$, and then $a \geq 0$ such that $p^a \geq N$. The elements 1_n and A commute, and all intermediate binomial coefficients for a p-power are divisible by p. Thus

$$\pi_n(g^{p^a}) = (1_n + A)^{p^a} = 1_n + A^{p^a} = 1_n.$$

The stable image of g^{p^a} therefore lies in $J(R)$. Its reduction to $\mathrm{St}(\mathrm{F}_p)$ is trivial, because g already belongs to the kernel of the fixed-rank reduction. By Theorem 14.1, some p-power p^b kills this stable element. Theorem 12.1 gives

$$g^{p^{a+b}} = 1 \quad \text{in } G_p.$$

Both exponents may depend on g. This proves the claimed torsion property and completes the proof. □

Corollary 4.1. *The groups G_p are separated by finite p-group images.*

Proof. Theorem 1.1 supplies finite separating images, and Lemma 5.3 shows that each such image has p-power order. □

5. Graded nil algebras and finite p-group images

Proposition 5.1. *Let I be an infinite-dimensional positively graded associative F_p-algebra with finite-dimensional homogeneous components. If every element of I is nilpotent, then $1 + I$ is an infinite residually finite p-group in the unitization $F_p \oplus I$.*

Proof. The identity $(1 + x)(1 + y) = 1 + (x + y + xy)$ gives closure under multiplication. If $x^N = 0$, the inverse of $1 + x$ is the finite series $\sum_{j=0}^{N-1} (-x)^j$. Thus $1 + I$ is a group. For $p^a \geq N$ one has $(1 + x)^{p^a} = 1$.

For $d \geq 1$, the quotient $I/I_{\geq d}$ is a finite-dimensional nilpotent algebra over F_p. Projection gives a homomorphism

$$1 + I \to 1 + (I/I_{\geq d}).$$

The target has cardinality $p^{\dim(I/I_{\geq d})}$. If $x \neq 0$, choose d above the degree of a nonzero homogeneous component of x. Then $1 + x$ is not killed by this homomorphism. These finite groups therefore separate all nonidentity elements. The map $x \mapsto 1 + x$ is a bijection of sets, so the group is infinite. □

Corollary 5.2. *If I is matrix-nil, the same conclusion holds for $1 + M_n(I)$, for each finite $n \geq 1$. Its degree-d quotient has cardinality $p^{n^2 \dim(I/I_{\geq d})}$.*

Proof. Apply Proposition 5.1 to the graded nil algebra $M_n(I)$. □

Neither result asserts finite generation or finite presentation of these full groups. The finite-presentation step of Theorem 1.1 is a separate assertion about a finite-index subgroup of a fixed-rank Steinberg group.

Lemma 5.3. *Every finite image of a p-group has order a power of p.*

Proof. Let F be a finite image. Every element of F has p-power order. Suppose a prime q divides $|F|$. Consider the tuples $(a_1, \ldots, a_q)$ with product 1. There are $|F|^{q-1}$ of them, a number divisible by q. Cyclic rotation preserves this set: moving the first factor to the end conjugates the total product. Since q is prime, every rotation orbit has size 1 or q. The number of fixed tuples is therefore divisible by q.

Fixed tuples have the form $(a, \ldots, a)$ with $a^q = 1$. The identity gives one fixed tuple, so there is a nonidentity such a. It has order q, which is possible in a p-group only when $q = p$. All prime divisors of $|F|$ are thus p. □

6. Complete symmetrization in characteristic p

Lemma 6.1. *Let K be a field of characteristic p, and let $X_1, \ldots, X_m \in M_d(K)$. If $m > (p-1)d^2$, then*

$$\sum_{\sigma \in S_m} X_{\sigma(1)} \cdots X_{\sigma(m)} = 0.$$

Proof. The expression is multilinear. Expand each argument in the d^2 matrix units. In each resulting tuple, at least p labeled arguments are equal, since $m > (p-1)d^2$. Fix p of these labels. Relabeling them defines a free action of S_p on the set of permutations of all m labels. Every product is constant on an orbit. Each orbit contributes $p!$ copies of the same product and therefore contributes zero. Sum over the orbits, then over the multilinear expansion. □

Lemma 6.2 (A free packet of chunks). *Suppose a finite collection of summands is partitioned into orbits, each orbit having contribution*

$$U \left(\sum_{\sigma \in S_m} X_{\sigma(1)} \cdots X_{\sigma(m)}\right) V \otimes w,$$

where the matrices have size d over a field of characteristic p, and $m > (p-1)d^2$ for each orbit. Then the total sum is zero.

Proof. Lemma 6.1 makes the contribution of every orbit zero. □

A useful sufficient size bound is obtained as follows. Suppose a run of length r contains complete chunks of length at most b, with at most two additional chunks crossing the ends. If there are m complete chunks, then $r \le mb + 2b$. Hence

$$r > b((p-1)d^2 + 2) \quad \Rightarrow \quad m > (p-1)d^2.$$

This numerical bound applies only after the summands have genuinely been partitioned into full permutation orbits with a common word factor w. It does not imply that an arbitrary word presentation has such orbits.

Lemma 6.3 (Absorbing a fixed prime). *Suppose $h_j = h_0 D^j$, with $h_0 > 0$ and $D > 1$, and a sequence $d_j \ge 1$ satisfies*

$$\log_2(d_j^2) \le A + Bj + \theta h_j, \quad \theta < 1.$$

For every fixed prime p and positive constant c, one eventually has $2^{h_j} > c((p-1)d_j^2 + 3)$.

Proof. The difference $h_j - \log_2(d_j^2)$ is at least $(1-\theta)h_0 D^j - A - Bj$, which tends to infinity. Thus $2^{h_j}/d_j^2$ tends to infinity. Since $d_j \ge 1$, the claimed inequality follows. □

7. Recovering matrix powers from word-length sums

Let A be a unital associative F_p-algebra generated by a finite alphabet Σ, and let I be the span of the images of nonempty words. Assume these images form a two-sided ideal. For a word v, write $[v]$ for its image in A and $|v|$ for its length. Linear dependence among word images is allowed.

Set $K = \mathrm{F}_p(z)$, where z is a central indeterminate. A finite chunk list consists of nonempty words v_γ, indexed by a finite set $\mathbb{C}$, and numerical weights $W_\gamma \in \mathrm{M}_d(K)$. Its length-$n$ sum is

$$C_n = \sum W_{\gamma_1} \cdots W_{\gamma_r} \otimes [v_{\gamma_1} \cdots v_{\gamma_r}],$$

where the sum runs over all sequences $\gamma_1, \ldots, \gamma_r$ with $\sum |v_{\gamma_i}| = n$. It is a finite sum because every chunk has positive length.

Proposition 7.1. *Suppose that, for every finite matrix size and every finite chunk list over K, the sums C_n vanish for all sufficiently large n. Then I is matrix-nil.*

Proof. Fix $T \in \mathrm{M}_d(I)$. Expand its entries into finitely many nonempty word images and collect the scalar matrix coefficients to write

$$T = \sum_{\gamma \in \mathbb{C}} A_\gamma \otimes [v_\gamma], \quad A_\gamma \in \mathrm{M}_d(\mathrm{F}_p).$$

If the list is empty then $T = 0$. Otherwise let b be the largest chunk length and apply the hypothesis to the weights $W_\gamma = zA_\gamma$. For each n, the corresponding sum is the image under scalar extension of the polynomial

$$P_n(z) = \sum z^r A_{\gamma_1} \cdots A_{\gamma_r} \otimes [v_{\gamma_1} \cdots v_{\gamma_r}],$$

with the same word-length constraint. For all $n \geq N$, its image in $\mathrm{M}_d(K \otimes A)$ is zero. The map $\mathrm{M}_d(A)[z] \to \mathrm{M}_d(K \otimes A)$ is injective: expand the finitely many coefficients in an F_p-basis of A, then use injectivity of $\mathrm{F}_p[z] \to \mathrm{F}_p(z)$ for each scalar coefficient. Therefore $P_n(z)$ is the zero polynomial whenever $n \geq N$.

Expanding T^a and grouping its products of exactly a chunks by total word length gives

$$T^a = \sum_{n=a}^{ab} [z^a] P_n(z).$$

For $a \geq N$, every polynomial on the right is zero. Hence $T^a = 0$. The matrix size and T were arbitrary. □

8. Prime-to-p periods and separation

Lemma 8.1. *For each prime p there is a constant $C_p \geq 1$ such that*

$$p^{v_p(2^m+1)} \leq C_p m \quad \text{for all } m \geq 1.$$

Proof. The assertion is immediate for $p = 2$. Suppose p is odd, and let f be the order of 2 in $\mathbb{F}_p^\times$. If f is odd, then -1 is not in the cyclic subgroup generated by 2, so p never divides $2^m + 1$. Otherwise set $e = f/2$. The element 2^e has order two and therefore equals -1 in $\mathbb{F}_p$. It follows that p divides $2^m + 1$ exactly when $m = et$ with t odd.

Put $c = v_p(2^e + 1)$. If $y \equiv -1 \pmod{p}$ and u is odd and coprime to p, the binomial expansion of $(-1 + (y + 1))^u$ shows $v_p(y^u + 1) = v_p(y + 1)$: its linear term has strictly smaller valuation than all higher terms. Raising to the pth power increases that valuation by one. Indeed the linear term has valuation $v_p(y + 1) + 1$; every other term has larger valuation, since p is odd. Factoring t into a p-power and a coprime odd part gives

$$v_p(2^{et} + 1) = c + v_p(t).$$

Consequently $p^{v_p(2^m+1)} \leq p^c t = (p^c/e)m$ in the divisible cases. Taking $C_p = \max(1, p^c/e)$ proves the assertion in all cases. □

Corollary 8.2. *Let m_j be positive integers tending to infinity and put*

$$L_j = 2^{m_j} + 1, \quad Q_j = \prod_{i=0}^{j} L_i, \quad \ell_j = Q_j / p^{v_p(Q_j)}.$$

Then $\gcd(\ell_j, p) = 1$ and ℓ_j tends to infinity.

Proof. The prime-to-p part of L_j divides ℓ_j. By Lemma 8.1 it is at least $(2^{m_j} + 1)/(C_p m_j)$, which tends to infinity. □

Lemma 8.3 (Reduction of phases). *Suppose long nonzero word classes have start and end phases modulo Q, with end phase equal to start phase plus length. Suppose their phases respect subword offsets, and a nonzero product has matching middle phases. If ℓ divides Q, reducing both phases modulo ℓ preserves all these properties.*

Proof. Reduction modulo ℓ is a well-defined homomorphism from $\mathbb{Z}/Q\mathbb{Z}$ to $\mathbb{Z}/\ell\mathbb{Z}$. Apply it to each phase identity. In particular, unequal middle phases after reduction exclude a nonzero product. □

9. A criterion for stable separation

Proposition 9.1. *Let R be a graded algebra with degree-zero part F_p. Suppose that every element z killed by every degree-truncation map on $\mathrm{St}(R)$ satisfies $z^{\ell} = 1$ for some positive integer ℓ coprime to p. Then the truncations separate $\mathrm{St}(R)$.*

Proof. Let z be killed by every map $\mathrm{St}(R) \to \mathrm{St}(R_d)$. Its matrix image involves only finitely many coordinates. Every entry of the difference between that matrix and the identity lies in $I_{\geq d}$ for every d. Since

$$\cap_{d \geq 1} I_{\geq d} = \{0\},$$

the matrix image of z is the identity. Thus $z \in J(R)$. The case $d = 1$ says that its reduction to $\mathrm{St}(F_p)$ is trivial. Theorem 14.1 gives $z^{p^a} = 1$ for some a. By the additional hypothesis, $z^{\ell} = 1$ with $\gcd(\ell, p) = 1$. Choose integers u, v such that $up^a + v\ell = 1$. Powers of the same element commute, and hence

$$z = (z^{p^a})^u (z^{\ell})^v = 1.$$

The intersection of the truncation kernels is therefore trivial. □

Corollary 8.2 supplies arbitrarily large integers coprime to p, and Lemma 8.3 allows compatible phases to be reduced to such moduli. Sections 19-22 establish the corresponding group identity $z^{\ell} = 1$ for the same Steinberg-group element and for the algebra constructed in Section 16. The group calculation is essential: the period estimate by itself would not imply this identity.

10. Finite presentation over a finitely presented algebra

Theorem 10.1. *If R is a finitely presented unital associative F_p-algebra and $n \geq 5$, then $St_n(R)$ is finitely presented as an abstract group.*

Write $T = \mathrm{F_p}\langle \Sigma \rangle$ and $R = T/(f_1, \ldots, f_s)$, with finite Σ and finitely many polynomial relators. Include the empty word ε among the monomials. Start with generators $g_{ij}(w)$ for $i \neq j$ and $|w| \leq 2$. Impose the following finite families:

$$g_{ij}(w)^{\mathrm{p}} = 1;$$

$$[g_{ij}(u), g_{kl}(v)] = 1$$
$$\text{if } |u|, |v| \leq 2, \quad j \neq k, \quad i \neq l;$$

$$[g_{ij}(u), g_{jk}(v)] = g_{ik}(uv)$$
$$\text{if } |u| + |v| \leq 2, \quad i, j, k \text{ distinct.}$$

Define a group word $\xi_{ij}(w)$ recursively. At length at most two it is the displayed generator. Otherwise write $w = av$ with first letter a, choose the least index h outside $\{i,j\}$, and set $\xi_{ij}(w) = [\xi_{ih}(a), \xi_{hj}(v)]$. For $f_t = \sum_w c_{tw} w$, add

$$\prod_w \xi_{ij}(w)^{c_{tw}} = 1 \quad (1 \leq t \leq s, \quad i \neq j).$$

The coefficients are represented by integers from 0 to $p - 1$, and the finite product is initially taken in any fixed monomial order. We prove that this presentation gives $\mathrm{St_n}(R)$.

A commutator identity used in the induction

Let $A = [X,Y]$ and $B = [Y,Z]$. Suppose A commutes with X,Y,B, B commutes with Y,Z, and X commutes with Z. Then $[A,Z] = [X,B]$, and this common element centralizes A,Z,X,B. Indeed, $\langle A,Z \rangle$ and $\langle X,B \rangle$ centralize one another. Conjugating $[X,Z] = 1$ by Y yields $[A^{-1},Z][X,B] = 1$. The two factors lie in the intersection of these commuting subgroups and are central there. Thus $[A^{-1},Z] = [A,Z]^{-1}$, proving the assertion.

10. Finite presentation (continued)

Induction on monomial length

Temporarily omit the polynomial relators and call the presented group P. At stage N construct roots $y_{ij}(w)$ for $|w| \le N$, proving simultaneously: independence of all positive splits and intermediate indices; order dividing p; all commuting-position relations between these roots; and every consecutive-root relation of total length at most N, including empty-word factors. The initial stage $N = 2$ is exactly the defining relations.

At a new length $N \ge 3$ consider $[y_{ih}(u), y_{hj}(v)]$ for every positive split $w = uv$. For three nonempty pieces u,v,z and four distinct indices i,a,b,j, the preceding identity gives

$$[y_{ib}(uv), y_{bj}(z)] = [y_{ia}(u), y_{aj}(vz)].$$

Every hypothesis of that identity concerns roots of smaller individual length, in commuting positions, and is already proved. Regard a candidate as a pair (cut position, intermediate index). Candidates with different cuts and different indices are equal by this formula. The resulting graph is connected: there are at least two cuts and at least three available intermediate indices. For equal cuts, pass through a different cut and a third index; for equal indices, use two other indices and alternate the two cuts. Hence all candidates agree and define $y_{ij}(w)$.

Next prove all commuting-position relations involving length N, by increasing total length from N to $2N$. For $[y_{ij}(u), y_{kl}(v)]$ with $|u| = N$, choose h outside $\{i,j,k,l\}$ and split $u = u_1u_2$ positively. Both factors of $[y_{ih}(u_1), y_{hj}(u_2)]$ commute with the other root by a smaller total length. Their commutator does also. This establishes the whole commuting family before any further new-length argument.

If the new root is $C = [U,V]$, these commutations say that C centralizes U,V. Since $U^p = 1$ at a smaller length, the identity $[U^p,V] = C^p$ gives $C^p = 1$. This argument works for every prime, with no involution assumption.

For an empty factor on the right, write $w = uv$ positively and choose h outside $\{i,j,k\}$. Apply the commutator identity to $X = y_{ih}(u)$, $Y = y_{hj}(v)$, $Z = y_{jk}(\varepsilon)$. The smaller-length relation for $[Y,Z]$ and positive-split independence give $[y_{ij}(w), y_{jk}(\varepsilon)] = y_{ik}(w)$. For an empty factor on the left use $X = y_{ij}(\varepsilon)$, $Y = y_{jh}(u)$, $Z = y_{hk}(v)$. This closes the induction.

10. Finite presentation (completion)

For a polynomial $f = \sum c_w w$, define $y_{ij}(f) = \prod y_{ij}(w)^{c_w}$. Same-position roots commute and have exponent dividing p, so this is additive in f over F_p. The other commuting relations extend factor by factor. For consecutive positions, every resulting monomial commutator commutes with both input products, so expansion gives

$$[y_{ij}(f), y_{jk}(g)] = y_{ik}(fg).$$

Thus these elements define $St_n(T) \to P$. The reverse map sends each original generator to its monomial root. Recursion and additivity show that the two maps fix all generators and are inverse.

Imposing the polynomial relators kills the roots of every element of their two-sided ideal. Indeed, if the roots of f_t are killed in every position, consecutive commutators first kill roots of af_t and then of af_tb. Additivity kills their finite sums. Consequently root parameters depend only on their class in R, and the quotient presentation is exactly $St_n(R)$. This proves Theorem 10.1. □

Theorem 18.1 supplies the finite algebra presentation needed here. Finite presentation is not inferred from grading or matrix nilpotence alone.

11. Local rings, strips, and centrality

In Sections 11-13 assume $B = \mathrm{F}_p \oplus K$, where K is an ideal and every finite matrix over K is nilpotent. A matrix is invertible precisely when its reduction over F_p is invertible: lift the inverse of the reduction and invert the remaining matrix $1 + A$ by its finite geometric series. In particular, the nonunits of B are exactly K. Independent reduced columns extend to an invertible matrix by lifting a complementary basis.

Put $H_n = \mathrm{GL}_n(B)$ and $J_n(B) = \ker \pi_n$. Row and column elimination using an invertible pivot shows that H_n is generated by elementary matrices and matrices $d_i(u)$ with a unit in one diagonal position. A literal row interchange is a product of elementary matrices and a diagonal sign. This observation keeps the argument valid when p is odd.

A presentation supported on at most five coordinates

Lemma 11.1. *For $n \geq 5$, H_n is presented by these generators and all true matrix relations supported on at most five coordinates.*

Proof. The complex of columns with independent reductions is simply connected for $n \geq 3$. To verify this, first take a finite labeled spanning collection over the field. Begin with a basis simplex and add labels. The link of an added vector is the independence complex of the preceding vectors projected modulo that vector, with zero projections discarded. Induction on rank gives vanishing reduced homology through rank minus two, using the relative chains of the attached cone. In rank at least three the links are connected. Attaching their cones preserves simple connectivity by replacing paths through a cone with paths in its link. Any loop in the full column complex lies in such a finite collection after standard basis columns have been added.

Let H'_n be the group in the claimed presentation. Induct on n, starting with $n = 5$, where every matrix relation is included. The stabilizer P of the first standard column is a row strip semidirect $\mathrm{GL}_{n-1}(B)$. By induction its complementary block lifts to H'_n; the strip and its conjugation laws involve at most three coordinates and also lift. Hence P has an injective section P'. Let s be the literal swap of the first two columns, expressed using the allowed elementary and diagonal generators. Then P' and s generate H'_n.

Transport a coset hP' along an edge from its column to a neighboring column by $hP' \mapsto h\tilde{p}sP'$, where $p \in P$ sends the second standard column to the prescribed neighbor in h-coordinates. The subgroup fixing the first two columns is generated by the remaining complementary block and the two row strips. The swap normalizes its lift by relations on at most three coordinates. Therefore transport is independent of the chosen p and of the coset representative.

Backtracking is trivial since $s^2 = 1$. A triangle can be carried to the first three standard columns. Writing t for their last-two swap, its successive representatives are s, sts, $ststs = t \in P'$; this is a three-coordinate matrix relation. Simple connectivity makes transport depend only on the endpoint. Every coset is reached because P' and s generate. Thus the cosets correspond bijectively to columns. An element in the kernel of $H'_n \to H_n$ lies in P', where the map is injective. □

11. Centrality and stable quotients

Lemma 11.2. *For $n \geq 6$ the kernel $J_n(B)$ is central. For $n \geq 5$ the quotients $GL_n(B)/E_n(B)$ are unchanged by stabilization.*

Proof. Disjoint sets of coordinates S,T determine an additive abelian strip generated by $x_{st}(a)$, $s \in S$, $t \in T$. Its matrix map is injective, since products within the strip have zero cross terms. Conjugation by roots supported on S or T acts by the exact corresponding row or inverse column operation. This follows from consecutive-root relations; the reversed consecutive commutator carries a minus sign.

Assign to an elementary matrix conjugation by its root, and to a diagonal matrix the automorphism $x_{ij}(a) \mapsto x_{ij}(u_i a u_j^{-1})$. These preserve root relations. Consider a true matrix relation supported on at most five coordinates, and choose one coordinate t outside them. Its action on the incoming and outgoing strips is respectively the action of its matrix and its inverse, so it fixes both strips. It fixes roots outside the support and, by $x_{ij}(a) = [x_{it}(a),x_{tj}(1)]$, also the roots inside. Lemma 11.1 therefore makes this an action of $GL_n(B)$ on $St_n(B)$. The action of $\pi_n(z)$ is conjugation by z, so an identity matrix image implies centrality.

Elementary and diagonal generators normalize $E_n(B)$, proving normality. To compute the quotient, kill the elementary generators in Lemma 11.1. For arbitrary characteristic the elementary signed interchange

$$e_{ij}(1)e_{ji}(-1)e_{ij}(1)$$
$$\text{has block} \quad ((0, 1), (-1, 0)).$$

conjugates $d_i(u)$ to $d_j(u)$; the signs are central scalars. Thus all diagonal positions have the same class $d(u)$. Diagonals in distinct positions commute. The remaining relations are exactly the true relations supported on at most five coordinates with elementary factors deleted and diagonal positions identified. This list is independent of $n \geq 5$, proving stabilization of the quotient. □

Lemma 11.3. *For every unital ring B, the stable matrix kernel $J(B)$ is central. Coordinate re-embeddings fix its elements.*

Proof. If a root word z has identity matrix image, take a spare coordinate outside its support. The faithful incoming and outgoing strips show that z commutes with all roots involving that coordinate. Express any internal root as a consecutive commutator through the spare coordinate. Repeat for any other external coordinate; disjoint roots already commute. Thus z is central. To re-embed its occupied coordinates, first transfer them one by one to a disjoint temporary set and then to the desired set. Each transfer uses an elementary signed interchange sending the occupied basis vector to an unused basis vector with coefficient +1. No root in the current word uses both interchanged coordinates, so the exact strip action proves the required conjugation on every root; the minus sign affects only the previously unused coordinate. Centrality fixes z throughout. □

12. Injectivity at a fixed rank

Theorem 12.1. *For B as in Section 11 and $n \geq 12$, the homomorphism $St_n(B) \to St(B)$ is injective.*

Central extensions of frame stabilizers

For $r \geq 6$ and $l \geq 0$ let $S_{r,l}$ be the matrices $((A,0),(C,1_l))$, with $A \in E_r(B)$ and arbitrary l by r strip C. Replace the elementary block by $St_r(B)$, acting through its matrix image, to obtain $\tilde{S}_{r,l}$. The map $\tilde{S}_{r,l} \to S_{r,l}$ is a central extension with kernel $J_r(B)$ by Lemma 11.2. Both groups are perfect: every internal or strip root is a consecutive commutator through an internal coordinate.

This is the universal central extension. Here is the root-lifting verification. In any central extension $V \to S_{r,l}$, choose lifts $y_{ij}(a)$ of all available roots, whose column index is internal. Lifts in commuting root positions commute exactly. Indeed, express one of their matrix images as a consecutive commutator through a spare internal coordinate. Each factor commutes with the other matrix root, so its commutator with a lift is central. Conjugation by the other lift fixes the commutator of those factors. That commutator differs from the chosen lift only by a central element.

Correct the lifts by setting $\hat{y}_{ij}(a) = [y_{ih}(a),y_{hj}(1)]$. The identity in Section 10, with an additional internal coordinate, gives

$$[y_{ih}(a), y_{hj}(b)] = \hat{y}_{ij}(ab).$$

To check this explicitly use $X = y_{ih}(a)$, $Y = y_{hf}(b)$, $Z = y_{fj}(1)$, replacing resulting root lifts by their central multiples. Taking $b = 1$ first proves independence of h. Expanding the corrected lift of $a + a'$ proves additivity, since the resulting commutators centralize the input factors. Thus the corrected roots satisfy all Steinberg relations and the exact action on the strip. They give a homomorphism $\tilde{S}_{r,l} \to V$. It is unique because its source is perfect and the difference between two lifts is a homomorphism to an abelian central kernel.

We use integral group homology, computed by the bar resolution. For a perfect group the kernel of its universal central extension identifies naturally with H_2. One algebraic description is this: if F/N is a free presentation, the extension $[F,F]/[F,N] \to F/N$ is universal and its kernel is $(N \cap [F,F])/[F,N]$, the degree-two homology group. The universal property follows by lifting free generators and observing that changes of lifts are central; the same kernel formula is obtained by resolving the free presentation in degrees one and two. In particular the block inclusion induces natural isomorphisms

$$H_2(E_r(B); \mathbb{Z}) \cong J_r(B) \cong H_2(S_{r,l}; \mathbb{Z}).$$

12. Ordered frames and their homology

Let $C_a(m)$ be the free abelian group on ordered independent $(a + 1)$-tuples of reduced columns of B^m, retaining the original columns as labels. Its differential deletes entries with alternating signs. Every ordering is a separate generator; no orientation relation identifies two orderings. Augment by $C_{-1} = \mathbb{Z}$.

Lemma 12.2. *This augmented complex is exact through degree $m - 2$.*

Proof. For a finite labeled spanning collection V over the field, induct on its number of labels, allowing rank zero. Delete a label v, obtaining V'. In the relative complex the tuples contain v. Filter by the number a of entries preceding v. For a fixed ordered prefix of length $a + 1$, the associated graded summand is the augmented ordered-frame complex of the remaining vectors projected modulo that prefix, shifted by $a + 1$. Their span has dimension $m - a - 1$. Induction gives vanishing through degree $(m - a - 3) + (a + 1) = m - 2$. If this quotient has dimension zero, its sole homology shifts to degree $m - 1$, outside that range. A finite filtration gives the same relative vanishing.

If deletion preserves rank, the long exact sequence and induction finish the proof. Otherwise v is independent of the span of V'. Prepending v is a chain homotopy killing the inclusion of its augmented complex. That inclusion is surjective on homology in the stated range by relative vanishing, hence the target homology is zero. Finally every cycle of the full column complex lies in a finite collection; adjoining the standard basis columns makes it spanning. □

The comparison of successive ranks

Fix $m \geq 13$. For frames of length $l \leq 5$, $\mathrm{E}_m(B)$ acts transitively. Complete the frame in the last l columns to an invertible matrix. Lemma 11.2 lets a complementary GL_{m-l} block cancel its class modulo E_m, while leaving the frame unchanged. Its stabilizer is $S_{m-l,l}$: the same quotient injectivity forces its complementary block to lie in E_{m-l}.

Take a free right $\mathbb{Z}[\mathrm{E}_m]$-resolution $P_\bullet \to \mathbb{Z}$ and the double complex $P_b \otimes C_a(m)$, with $a,b \geq 0$. Adding the augmentation column and using Lemma 12.2 shows that augmentation identifies total homology in degree two with $\mathrm{H}_2(\mathrm{E}_m;\mathbb{Z})$. Taking vertical homology first gives, for $0 \leq a \leq 4$,

$$E^1_{a,b} = \mathrm{H}_b(S_{m-a-1,a+1};\, \mathbb{Z}).$$

This identification follows directly because the permutation module of a transitive frame set is the module induced from its stabilizer; restricting the free resolution computes that stabilizer’s homology. The first differential is the alternating sum of the face inclusions, after returning each face to the standard frame.

12. Completion of the stability argument

The row $b = 1$ vanishes in these columns, since the stabilizers are perfect. The row $b = 0$ consists of $\mathbb{Z}$ with every face map the identity, so its differential is the identity for positive even a and zero for odd a. Consequently

$$E^2_{1,1} = E^2_{2,1} = E^2_{2,0} = E^2_{3,0} = 0.$$

In row $b = 2$, the two faces from column one to column zero induce the same map. By the central-extension computation it suffices to compare them on the first $m - 2$ coordinates. One face already gives the last standard column. To return the other to it, use the elementary determinant-one matrix on the last two coordinates sending e_{m-1} to e_m and e_m to $-e_{m-1}$. This fixes the entire complementary block. Thus the two face maps coincide in every characteristic, and their alternating sum is zero.

$$E^2_{0,2} = \mathrm{H}_2(S_{m-1,1}; \mathbb{Z}).$$

No higher differential leaves (0,2). The only possible incoming sources are (2,1) on page two and (3,0) on page three, both zero by the displayed vanishing. All other positions of total degree two vanish as well. The filtration of total homology therefore identifies the edge map with an isomorphism

$$\mathrm{H}_2(S_{m-1,1}; \mathbb{Z}) \to \mathrm{H}_2(\mathrm{E}_m(B); \mathbb{Z}).$$

The edge map is induced by inclusion: its zero-th column corresponds to the last standard column, and augmentation is the stabilizer inclusion on the free resolutions. Combining with the natural central-extension identifications proves that $J_{m-1}(B) \to J_m(B)$ is an isomorphism. Finally, an element killed by stabilization already has identity matrix image, since adjoining identity blocks is injective on matrices. It belongs to J_n and cannot be killed at any later rank when $n \geq 12$. The direct limit detects equality at a finite stage. This proves Theorem 12.1. □

13. Finiteness of the truncated Steinberg groups

Lemma 13.1. *A finitely generated perfect group with a central subgroup of finite index is finite.*

Proof. Let Z be the central subgroup and let its index be q. A finite coset transversal rewrites generators of the ambient group into finitely many generators of Z, by the same telescoping argument as in Section 3; finite presentation is unnecessary here. Thus Z is a finitely generated abelian group.

Choose representatives $t_1, \ldots, t_q$ for its cosets. For an element g, write uniquely $t_i g = z_i(g) t_{\sigma g(i)}$, where $z_i(g) \in Z$ and σ_g is a permutation. Multiplication and centrality give a homomorphism

$$V(g) = \prod_{i=1}^{q} z_i(g) \quad \text{from the group to } Z.$$

For $z \in Z$ it satisfies $V(z) = z^q$. Perfectness makes every homomorphism to an abelian group trivial, so every element of Z has order dividing q. If Z has r generators it has at most q^r elements. Finite index now makes the original group finite. □

Theorem 13.2. *For a finite algebra $B = F_p \oplus K$ with matrix-nil K and $n \geq 6$, the group $St_n(B)$ is finite.*

Proof. There are finitely many root generators since B is finite. The group is perfect by Lemma 2.1, and its kernel over $E_n(B)$ is central by Lemma 11.2. The matrix group is finite, so this kernel has finite index. Apply Lemma 13.1. □

Corollary 13.3. *Let $R = F_p \oplus I$ be positively graded with finite-dimensional homogeneous parts. For $n \geq 6$, every $St_n(R_d)$ is finite.*

Proof. Its underlying vector space is the sum of the finitely many finite-dimensional components of degrees 0 through $d - 1$. Its positive ideal has dth power zero. Any matrix over that ideal has dth power zero too, since each entry is a sum of products of d ideal elements. Theorem 13.2 applies, including $d = 1$. □

In particular, all truncation targets used at rank twelve are finite. The central-kernel argument is necessary in addition to finiteness of the ring.

14. Relative p-power torsion in the stable kernel

In this section B is any unital F_p-algebra. Use outer indices i,j and inner indices $0 \le s,t < b$. For a matrix $A = (a_{st})$ define the block root

$$X_{ij}(A) = \prod_{s,t} \mathrm{x}_{(i,s),(j,t)}(a_{st}) \quad (i \neq j).$$

Its factors commute. Consecutive block commutators multiply matrices because the only nontrivial constituent commutators have a common middle coordinate and all their output roots centralize both blocks. Together with addition and commuting positions this defines $\mathrm{St}_n(\mathrm{M}_b(B)) \to \mathrm{St}_{nb}(B)$.

A numerical basis change $P \in \mathrm{GL}_b(\mathrm{F}_p)$ can be implemented simultaneously in all outer blocks in the stable group. Adjoin a spare coordinate and compensate the determinant of $\mathrm{diag}(P, \ldots, P)$ there. Factor each P by elimination into transvections and diagonal scalings. Implement a scaling u with its inverse on the spare coordinate: if $w(u) = \mathrm{e}_{12}(u)\mathrm{e}_{21}(-u^{-1})\mathrm{e}_{12}(u)$, direct multiplication gives $w(u)w(-1) = \mathrm{diag}(u,u^{-1})$. The strip actions verify at each step that the lifts conjugate block roots to $X_{ij}(P^{-1}AP)$. This supplies exact group conjugation in every characteristic, including determinant compensation.

Triangular substitution

Let $v(t)$ be a root word over $B[t]$, where t is central, with identity matrix image. A numerical matrix T may be substituted for t; write $W_v(T)$ for the result followed by block roots. If T is upper triangular with constant diagonal λ, then

$$W_v(T) = \prod_{s=0}^{b-1} \iota_s(v(\lambda)),$$

where ι_s puts a root in the coordinates with inner index s. To prove this, consider the group generated by all roots of strictly positive inner gap $t - s$, allowing the outer indices to agree. Commutators add gaps, and opposite positions cannot occur. Collecting by increasing gap gives one coefficient at each position. If its matrix image is one, entries at the least remaining gap are exactly those coefficients, so they vanish successively. This group is faithful in matrices. Roots of gap zero normalize it, as every resulting commutator has positive gap. Split each evaluated block root into its diagonal part and positive-gap part, and collect. The remaining positive-gap factor has identity matrix image and is therefore trivial. This proves the formula.

14. Removing the wrap of a cyclic shift

Let C_b be the shift with entries 1 from row s to column $s + 1$ modulo b, and let S_b be the same shift with its final wrap entry deleted. Suppose $v(t)$ has m factors and all polynomial entries have degree at most d. Put $\delta = \max(1,d)$. We prove

$$b > 2(m + 1)\delta \quad \Rightarrow \quad W_v(C_b) = W_v(S_b).$$

Split each cyclic block root into $D_h F_h$, where D_h is its truncated evaluation and F_h consists of wrap entries. A degree at most d term wraps at most once, so every correction goes from the last δ inner coordinates to the first δ inner coordinates. These form a faithful abelian strip.

For $r \geq 1$ let V_r be the strip from the last $r\delta$ inner coordinates to the first $r\delta$ inner coordinates, with all outer coordinates included. These sets remain disjoint up to $r = m + 1$. Conjugation by one D_h sends V_r into V_{r+1}. Indeed its constituent roots all have the same distinct outer source and target, and forward inner displacement between zero and d. A commutator can move the source endpoint left by at most d, or the target right by at most d. After a source move the new outer source cannot match another constituent's target; after a target move the new outer target cannot match another constituent's source. Each endpoint moves at most once.

There is no opposite-root interaction: initially the backward separation exceeds $b - 2r\delta$, and after one endpoint move it still exceeds $b - (2r + 1)\delta > \delta$. This is greater than or equal to every forward displacement. A root after both moves commutes with all constituents. Thus all corrections stay in the enlarged strip, with the appropriate signs in reverse commutators.

Put $D_{>h} = D_{h+1} \cdots D_m$. The exact collection identity is

$$\prod_{h=1}^{m} (D_h F_h)$$
$$= (D_1 \cdots D_m) \prod_{h=1}^{m} (D_{>h}^{-1} F_h D_{>h}).$$

Every final correction belongs to V_{m+1}. Therefore $W_v(C_b) = W_v(S_b)u$ with u in a faithful strip. Both evaluations have identity matrix image, so u does also and equals one. This proves the claimed identity.

14. The relative torsion theorem

Theorem 14.1. *Let $R = \oplus_{a\ge 0} R_a$ be a nonnegatively graded unital F_p-algebra with $R_0 = F_p$. Every element of*

$$J(R) \cap \ker[\mathrm{St}(R) \to \mathrm{St}(\mathrm{F}_p)\,]$$

has finite p-power order. Neither matrix nilpotence nor finite presentation is needed for this assertion.

Proof. The homogeneous substitution $\sum r_a \mapsto \sum r_a t^a$ is a unital ring homomorphism $R \to R[t]$. For a fixed finite root word representing z in the displayed kernel it gives a polynomial root word $z(t)$ with identity matrix image. It evaluates to z at 1 and to its degree-zero reduction at 0. Substitution and block roots define $W_z(T)$ on the stable group. Different embeddings of its finite coordinates give the same central element by Lemma 11.3.

For all sufficiently large integers b, the wrap-removal identity followed by triangular substitution gives

$$W_z(C_b) = W_z(S_b)$$
$$= \prod_s \iota_s(z(0)) = 1.$$

The equality $z(0) = 1$ is in the stable group; each coordinate embedding is a homomorphism there, so no unstated fixed-rank injectivity is used at this step.

Now choose $b = p^a$ above that threshold. Since $C_b{}^b = 1$ and the characteristic is p,

$$(C_b - 1)^b = C_b{}^b - 1 = 0.$$

A basis adapted to the kernels of this nilpotent operator makes C_b upper triangular with every diagonal entry 1. The compensated numerical basis change described above implements this inside the stable Steinberg group. Both evaluated words belong to its central kernel, so conjugation leaves them equal. Triangular substitution and Lemma 11.3 now give

$$W_z(C_b) = \prod_s \iota_s(z) = z^b.$$

The previous vanishing therefore proves $z^{p^a} = 1$. □

Theorems 10.1, 12.1, and 14.1 and Corollary 13.3 provide the four general Steinberg-group steps in Theorem 1.1. Sections 16-22 construct the required algebra and prove stable truncation separation for it.

15. Why an algebra construction is indispensable

Proposition 15.1. *For every prime p there is an infinite-dimensional graded algebra $R = F_p \oplus I$, with finite-dimensional homogeneous components and matrix-nil I, for which $St_n(R)$ is not even finitely generated for any $n \geq 3$.*

Proof. Take $I = \oplus_{r\geq 1} F_p e_r$, put e_r in degree r, and define every product in I to be zero. Its unitization is an associative graded algebra with one-dimensional positive homogeneous components. Every matrix A over I satisfies $A^2 = 0$.

Any finite set of elementary matrices over R uses parameters involving only finitely many basis elements e_r. Their span V gives a finite subalgebra $F_p \oplus V$, closed under multiplication because $V^2 = 0$. A finite set of arbitrary elements of $E_n(R)$ can first be written as finitely many elementary factors, so it too belongs to $GL_n(F_p \oplus V)$, a finite group. Thus $E_n(R)$ is locally finite. It is infinite because the matrices $1 + e_r E_{12}$ are distinct. An infinite locally finite group is not finitely generated. Its surjective preimage $St_n(R)$ cannot be finitely generated either. □

This example shows why grading, infinite dimension, finite homogeneous pieces, and matrix nilpotence alone do not imply finite presentation of the group. Sections 16-18 construct an algebra with an ordinary finite presentation as well as all of these properties.

Truncations always separate matrices over a graded algebra, but the group argument must also handle elements of $J(R)$. Theorem 14.1 gives relative elements p-power order. Sections 19-22 use the additional structure of the constructed algebra to obtain an annihilating exponent coprime to p for elements killed by every truncation. Proposition 9.1 then gives separation.

16. A finite hierarchy of word rules

A word presentation consists of replacements $u \leftrightarrow v$ of equal positive length and declarations $w = 0$. A replacement may be used inside any word, in either direction. Over a field K the corresponding algebra is the free unital associative algebra modulo the two-sided ideal generated by $u - v$ and the declared zero words.

Lemma 16.1 (Word basis). *The nonzero replacement classes form a basis over any field. A word is zero exactly when a finite sequence of replacements reaches a word containing a declared zero factor.*

Proof. Form the graph on all words with contextual replacements as edges. The span of edge differences is the span of all differences within a connected component. The quotient therefore has one basis vector per component. Multiplication concatenates components, so imposing the zero declarations kills exactly the components meeting one of their contexts. All rules preserve length, and none affects the empty word. □

Fix $c = 10000$, $k = 10$ and $D = 2^{21}$. In particular $D > 10(c + 1)(k + 1)$. The construction below uses no field operations. For all sufficiently large integer parameters h, put $q = 2^h$ and $L_h = 2q + 1$.

Theorem 16.2 (Hierarchy). *There are an integer h_0 and finite word presentations $A_h(K)$, for $h = h_0 D^j$, $j \geq 0$, with the following properties, over every field K.*

(a) *The alphabet Σ_h has size 2^{ch} and every defining rule has length at most k. A substitution E_h encodes an upper letter as a word of length L_h, induces $A_{Dh} \to A_h$, and preserves nonzero words. Each level has a nonzero letter.*

(b) *A subset of letters defines cuts immediately before their occurrences. Every sufficiently long nonzero word decomposes uniquely into a bounded prefix, the intervals between its consecutive cuts, and a bounded suffix. Each such interval t can be changed internally to $E_h(u_t)$, where $1 \leq |u_t| \leq k$, so $|t| = L_h|u_t|$. The token list and the bounds are finite.*

(c) *There are q palette colors. Every sufficiently long nonzero word contains a run of at least $q/100$ palette letters with identical non-color fields and no control heads. Within every such run, adjacent letters may be transposed and all colors are distinct. Its eligibility and maximal endpoints depend only on the non-color fields.*

The following pages specify the finite membership predicate for the rule table and prove these assertions. Enumeration of its finite input set gives an ordinary finite presentation at any chosen level; relations from other levels are never added as extra relators.

16. Alphabet and structural blocks

Let Σ_h be all bit strings of length ch. Reserve 32 integer fields of width $2h$ and 512 one-bit fields, using fixed disjoint slots. Name the fields as needed below, in the order of their introduction; fill all unused slots with zero. Integers use binary notation. Finite tags use the initial available flag slots. Every out-of-range value or unlisted combination is a unary zero. The listed fields occupy fewer than these fixed allowances. Large constants occurring in the computation program are encoded in data, not in the alphabet format.

A letter has one structural type: $C(r)$ with $1 \le r \le q$, a palette letter $P(a)$ with $1 \le a \le q$, or a stop S. The permitted neighboring types are precisely

$$C(r)C(r+1), \quad C(q)P(a), \quad P(a)P(b),$$
$$P(a)S, \quad SC(1).$$

Here the first type requires $r < q$; all unlisted pairs are zero. A full structural block consists of q core cells, a nonempty palette, and one stop. No test is made for a neighbor absent at an endpoint of the whole word.

Each letter has a transaction track. Each palette letter also has a checker track; on other structural types the checker fields are idle and empty. An idle core cell has a data bit at positions $r \le cDh$, and no data bit at other positions. We will choose h large enough that $cDh \le q$. Palette and stop cells have no data bits. A canonical block has both tracks idle, each palette color once in increasing order, and its cDh core data bits equal to an upper letter in Σ_{Dh}. This defines E_h on every upper bit string, even those later declared zero. Its length is $L_h = 2q + 1$.

A nonidle transaction cell has a marker L, R, B, or F: respectively a left wing, right wing, acquisition cursor, or verification head. It carries a proposed number of blocks $1 \le l \le k$, a task tag “replacement” or “zero”, and a relative block number $1 \le b \le l$. Only a cursor or full head carries the selected data side $\varepsilon \in \{0,1\}$; a zero task always has $\varepsilon = 0$. Only a full head carries the verifier control and three reading bits.

A nonidle core cell has two data slots at each position $r \le cDh$, and one certificate bit at every core position. Other cells have no payload. A nonidle cell is at its known first position when $b = 1$ and its type is $C(1)$; it is at its known last position when $b = l$ and its type is S. An R at a known first position and an L at a known last position are unary zeros. These tests apply even when a whole-word neighbor is absent.

16. The local interface rules

A side of a transaction cell has an internal or external interface. The complete list is as follows. “First” and “last” refer to the known positions just defined.

Idle. Both interfaces are external.

Left wing L. The left interface is external at first, otherwise internal from an *L*. The right interface is internal to *L*, *B*, or *F*.

Cursor B. The left interface is external at first, otherwise internal from an *L*. The right interface is external.

Full head F. The left interface is external at first, otherwise internal from an *L*. The right interface is external at last, otherwise internal to an *R*.

Right wing R. The left interface is internal from *F* or *R*. The right interface is external at last, otherwise internal to an *R*.

Facing interfaces must both be external, or both be internal with the prescribed markers. An internal pair has identical task and *l*; its relative block number is unchanged within a block and increases by one across *SC*(1). It cannot enter a known first position or leave a known last position. Every pair also satisfies the structural test. These are all transaction pair tests; any violation is a binary zero.

The checker on a palette has states idle, passed(x), or head(x), for a sought color $1 \le x \le q$. Its permitted neighboring states are exactly

$$\text{(idle,idle)},\quad \text{(passed}(x)\text{,passed}(x)\text{)},$$
$$\text{(passed}(x)\text{,head}(x)\text{)},\quad \text{(head}(x)\text{,idle)}.$$

At the first palette cell any of these states is permitted. At its last cell only idle or head is permitted, tested by its pair with *S*. The tests at the preceding *C*(*q*) and at *S* are binary. Thus a complete palette is either idle or a passed prefix, one checker head, and an idle suffix. Partial versions are allowed at endpoints of the whole word.

A complete acquired region consequently has the form *L*FR** on exactly its *l* blocks. A partially acquired region is *L*B*, possibly ending inside a block, with the rest of that block idle. Adjacent regions may have different metadata at an external joint. These consequences follow from the pair tests; no assumption that a word arose by a computation is imposed.

16. Palette moves and acquisition

All replacements preserve every field not explicitly changed, and include their inverses. At $C(q)$ followed by an idle-checker palette cell, a checker head seeking any x may be created. On neighboring palette cells, (head(x),idle) may become (passed(x),head(x)) if the first cell's color is not x. Finally two neighboring palette colors may be swapped when their other fields agree and neither cell carries either kind of head. Because the other fields agree, this is also a swap of whole letters.

Declare zero a passed(x) cell of color x, any two adjacent palette cells of equal color, and a final palette head seeking x whose color differs from x. No other palette moves or palette zeros are used.

Palette invariant. Starting with each color once, no palette zero can be reached. Colors are preserved as a multiset. A checker cannot pass its sought color. Swaps cannot cross its head or the boundary between different checker states, so a passed prefix continues to exclude that color. A head therefore cannot finish incorrectly. Equal neighboring colors are impossible. Transaction moves, specified below, change neither colors nor checker fields.

Conversely, in a nonzero complete palette the checker can be removed. If a passed cell has the sought color there is already a zero; otherwise reverse its steps until its head reaches the first cell and reverse creation. Once the transaction track is idle too, all non-color fields agree. Repeated colors can then be swapped together to produce zero; a missing color can be sought all the way to the final failed check. Thus a nonzero idle palette has every color exactly once and can be sorted.

Acquisition begins by changing an idle $C(1)$ to a cursor with $b = 1$ and chosen l, task and ε. Copy its idle data into slot ε and freely choose its other slot and certificate bit. A cursor may acquire its next cell only if that cell is transaction-idle and is the next internal position of its intended region. The old cursor becomes L, the next cell becomes the cursor, and its own structural position and relative block number determine the transported metadata. Existing core data are copied to slot ε; the other data and certificate bits are arbitrary. No payload is placed on a palette or stop.

The cursor cannot advance beyond its known last position. At that position alone it may turn into a full head in the initial verifier phase with all reading bits zero. All these moves have exact inverses. Inverse acquisition restores the selected slot to the idle data field; inverse creation removes the cursor. Acquisition and release are the only moves which can change stored payload.

16. A reversible verifier for a finite clause list

Temporarily fix a finite program string a. The next two pages construct from a,h,l and the task a nonempty list $\Phi_1, \ldots, \Phi_m$, with $m \le q$. Each clause reads at most three bits: either a specified data slot at a specified relative block and core position, or a certificate bit in the first block. Pad shorter clauses by repeated addresses or an ignored dummy address. Both data slots are read for replacements; only slot zero is read for zero tasks. The clauses do not depend on ε.

For each clause j the full-head path is

$$A_j \leftrightarrow \text{reading tour} \leftrightarrow \text{before-gate}_j$$
$$\leftrightarrow \text{after-gate}_j \leftrightarrow \text{undo tour} \leftrightarrow Z_j.$$

Its endpoint phases occur only at the known last position. The head carries the clause index, one of these finitely many phases or the directional tour phases, ε, and three reading bits ρ. Every tuple ρ is syntactically permitted. A phase A_j, Z_j, or success with $\rho \neq 0$ is a semantic unary zero. Such states remain syntactically present and can be endpoints of replacement edges. This distinction is part of the definition.

The reading tour starts at the last position, travels left, turns at the first, and returns right. A left step changes (L,F) to (F,R), and XORs into each reading register the addressed bit at the new head position, if any. A right step changes (F,R) to (L,F) and reads nothing. All payload and relative block numbers stay on their original cells; only head control, index, ε and reading registers move. The last position is a stop, never an address, so every addressed bit is read exactly once. Each turn, entry and exit is a unary phase change at its specified endpoint; each move has its exact inverse.

The gate from before-gate to after-gate exists exactly when $\Phi_j(\rho)$ is true. A false after-gate state is zero. A false before-gate state has no gate edge but is not thereby zero. The undo tour travels left without reading, turns at first, and returns right, XORing addressed bits when leaving their cells. It exits at last into Z_j. Its phases are distinct from the reading-tour phases, and its inverse edges are included.

All tour entry and exit edges preserve every register tuple, even a tuple making an endpoint a semantic zero. Only the acquisition interface, the edge $Z_j \leftrightarrow A_{j+1}$, and the final edge $Z_m \leftrightarrow$ success require zero registers. A replacement success with zero registers may flip ε; a zero-task success is a unary zero. Success has no shortcut back to acquisition. No phase is stored on the wings. This completely lists the verifier moves.

16. Correctness and rollback of the local protocol

Each listed move is instantiated on all unaffected fields allowed by the unary formats, with the indicated internal pair conditions for moves inside a region. In acquisition, test the changed pair as internal after the move. All moves and all zeros are unary or binary. There are no table entries of lengths three through *k*. Semantic zeros do not delete an otherwise listed replacement edge.

Context and ownership. Starting with complete canonical blocks, all syntactic pair rules survive every move. A new cursor at idle *C*(1) has an external predecessor and an idle successor inside the block. Acquisition changes the old cursor’s right interface from external to internal with its acquired neighbor; the new cursor has the external interface previously supplied by the idle cell. Its predecessor still permits an *L* or head with the same metadata. The known-last condition prevents an illegal last *L*. Release reverses these checks. A full-head move changes *F,R* to *L,F* or conversely, with exactly the same outside allowed interfaces. The endpoint change $B \leftrightarrow F$ has identical external right interfaces. Checker moves preserve their stated grammar, and neither kind of move changes the other track.

Every acquired portion of a block contains its *C*(1), or continues a region which previously acquired that cell. A new transaction can start only at an idle *C*(1), and acquisition only takes idle cells. Thus two transactions cannot own cells in the same block. A region may be blocked by a later region at an external joint, but cannot acquire its cells until they are released. This proves the context and ownership assertions also when regions abut.

Verification from acquisition. While a region is full, all its payload is fixed, even if moves elsewhere are interleaved. Starting at A_j with $\rho = 0$, the reading tour puts exactly the addressed bits in the registers. The gate is crossed precisely when the clause is true, and the undo tour restores $\rho = 0$. An incorrect clause breaks the path before its after-gate state. Backtracking cannot bypass it because the only clause changes and acquisition interface are the listed ones. Thus success is reachable from acquisition exactly when every clause holds. No register or gate zero can be reached on such a path. A side flip changes no tested bit, so the path may then be fully reversed.

Rollback from an arbitrary full state. Every tour edge is a bijection on the register tuple for fixed payload. Reverse the prescribed path to its preceding endpoint. A reversed reading tour reaches A_j; nonzero registers there expose a zero. A reversed undo tour reaches after-gate; a false value exposes a zero, and a true one crosses the gate backwards and reverses the reading tour. A false before-gate state can also reverse its reading tour. At A_j with zero registers reverse the previous interclause edge and decrease the clause index. Success first reverses to Z_m. This reaches acquisition or a zero in finitely many reversals. At acquisition, inverse acquisition releases every acquired cell and inverse creation finishes. A cursor region can immediately be released. All these operations stay inside that region.

16. Explicit computation certificates

Fix a deterministic universal one-tape machine with fixed finite symbol and state alphabets. It interprets a supplied binary program, returns to a marked origin when accepting, and then freezes in a fixed accepting configuration. Such a machine can simulate each fixed program with polynomial overhead: encode its tape tracks and head marks and scan the used segment to implement each step; that segment grows at most linearly with elapsed simulated time. The return to the origin adds at most the segment length. Fix this machine and its finite transition function once and for all.

$$T = 2^{\lfloor h/5 \rfloor}, \quad W = 4T, \quad o = 2T.$$

Use time rows $0 \le t < T$, tape cells $0 \le j < W$, and origin o. A tape cell, including a possible head state and the origin mark, has β bits, for an absolute constant β. The initial tape contains a self-delimiting encoding of the program a, parameter Dh, arity l, task, and the $lcDh$ bits from slot zero, followed by those from slot one for replacements. For example encode an integer n as $1^r 0\,\mathrm{bin}(n)$, where r is its binary length and $\mathrm{bin}(0) = 0$. Store this input immediately to the right of the origin; its size is $O(h)$ for fixed a,D. For large h it occupies less than $T/4$ cells. A head cannot reach either boundary in T steps.

A machine step has a radius-one Boolean rule on the three adjacent cell encodings. Extend it to a total function on all bit patterns and fix a NAND circuit with g gates computing its β outputs. The constants β and g do not depend on a,D,h. Invalid patterns do not occur in a satisfying tableau because the initial row is valid and the next row is forced.

Allocate certificate bits $X(t,j,r)$ for all cell bits and $Y(t,j,v)$ for gate outputs, at these one-based core addresses:

$$X(t,j,r) : 1 + \beta(Wt + j) + r,$$
$$0 \le r < \beta;$$

$$Y(t,j,v) : 1 + \beta WT + g((W-2)t + j - 1) + v,$$
$$0 \le t < T-1, \quad 1 \le j < W-1, \quad 0 \le v < g.$$

There are $\beta WT + g(W-2)(T-1)$ certificate bits. They are stored in the first relative block only. The program string is part of the prescribed initial tape, not a control-head field.

16. Clauses and a finite rule-deciding program

List four clause families in this order, with their internal indices in lexicographic order. First, constrain every initial-row bit, using two clauses for equality to a data bit or its negation, or a repeated unit clause for a constant. Second, constrain both boundary cells at every later time to remain blank. Third, for every interior cell and nonfinal time constrain all NAND gates and then all next-row output bits. A gate $z = \neg(u \wedge v)$ is expressed by

$$(u \vee z), \quad (v \vee z), \quad (\neg u \vee \neg v \vee \neg z).$$

Each output equality uses two clauses. Fourth, use β unit clauses fixing the final cell at the marked origin to its accepting head state and fixed origin symbol. Constants on circuit wires are permitted and can be ignored as inputs of the resulting Boolean clause. The total clause count is

$$2\beta W + 2\beta(T-1)$$
$$+ (3g+2\beta)(W-2)(T-1) + \beta.$$

Both this number and the certificate count are $O(T^2)$ and thus at most $q = 2^h$ for all large h. Every clause has at most three addressed inputs. Induction on time shows that a satisfying assignment is precisely the actual computation and its circuit outputs. The boundary conditions do not interfere with that computation. The final constraint holds exactly when acceptance occurs before T, since earlier acceptance freezes. This proves both directions of the certificate test.

Given one clause index, comparisons with the four family sizes, followed by quotient and remainder, find its family, time, cell and bounded gate subindex. The displayed address formulas and the fixed circuit then produce its addresses and truth function. Initial-row lookup chooses a constant bit, a program bit, a parameter bit or an addressed data bit. All arithmetic is on $O(h)$-bit integers and takes polynomial time in h for fixed a,D. One never enumerates the whole tableau or the preceding clauses.

Define $P(a,h;$ candidate) to decide whether a proposed unary or binary replacement or zero is in the table specified so far, using the clauses just defined with program string a. For replacements test both orientations. Check only the relevant formats, changed fields, interfaces, and at most one indexed clause. Include illegal formats as zeros, but do not suppress the listed moves with semantic-zero endpoints. All other candidates, including lengths three through k, are rejected. If any layout size, input length or field range does not fit, instead use the finite table declaring every letter zero and having no replacements. Its fit tests are explicit integer inequalities. The predicate P is polynomial-time for fixed a,D; it uses a as finite data and never executes it or searches for a certificate.

16. Closing the self-reference and choosing the parameter

There is an effective finite code constructor $s(t,v)$ whose output, on input x, simulates the two-input program t on (v,x). For example its code consists of the stored strings t,v and a fixed interpreter wrapper. Simulation by scanning has polynomial overhead for each fixed stored program.

Take the two-input program t which forms the finite string $s(v,v)$ from its first input v and computes $P(s(v,v),x)$ on its remaining input. Set

$$a = s(t,t).$$

Running this particular a on x returns exactly $P(a,x)$. Constructing the fixed string $s(t,t)$ costs a constant, while P and the interpreter overhead are polynomial in the parameter. This is a finite program definition. It is not a call by P to itself: P only checks local formats and the description of one Boolean clause.

The value D was fixed before constructing this code. Therefore the computation of a on each candidate at parameter Dh, including interpretation and return to the origin, has a polynomial time bound in h. Choose h_0 so large that all format and size bounds hold, $q \geq 4$, and that bound is less than $2^{\lfloor h/5 \rfloor}$ for every $h \geq h_0$. Exponential growth supplies such an integer; a polynomial upper bound for the explicitly described routines also gives an effective choice. At these parameters an accepting certificate exists exactly for a genuine table entry at Dh.

All alphabet control fields fit uniformly. The structural position, color, sought color, block count, relative block number and clause index use fewer than 32 integer slots of width $2h$; the largest indexed clause and certificate address is at most 2^h. Tags, phase, selected side, slots and registers use fewer than 512 bits. Tape encodings and gates consume certificate positions and are not alphabet tags. The code length and the fixed value D are accommodated by taking h_0 large; they do not enlarge c or k.

For each $h = h_0 D^j$, enumerate all unary and binary candidates on Σ_h and keep those for which this fixed P returns one. The resulting alphabet and relation lists are finite. Every statement below concerns that finite table alone.

16. Preservation of nonzero encoded words

A genuine upper replacement on l letters can be simulated inside its l canonical blocks. Acquire them with their old data in slot zero, the proposed new data in slot one, and a satisfying certificate in the first core. The verified path reaches success, flips ε, reverses all verification, and releases the cells. This leaves the target encoding. A genuine upper zero is simulated in the same way, reaching its declared zero-task success. No exterior cell is used. Thus E_{h} carries every upper relation to a consequence of the lower relations, and defines an algebra homomorphism.

For nonzero preservation, start with the encoding of an arbitrary upper word and follow an arbitrary sequence of lower replacements. The context and palette invariants preserve structural and marker syntax, and each palette continues to contain every color once. At every moment read a represented upper word: an idle core contributes its idle data; an acquired core contributes the slot selected by its unique owning head. The ownership property makes this unambiguous. Acquisition and release preserve those data, even when unselected slots and certificates are freely changed after partial release. Checker and palette swaps do not change the represented word.

During every interval in which a head is full, its payload is fixed and its control path began at acquisition. The verification argument therefore applies even when other regions move. A replacement side flip occurs only with a certificate of a genuine upper replacement, so it changes the represented word by that replacement. A zero-task success would certify a genuine upper zero. Syntactic zeros, false-after-gate zeros and nonblank endpoint zeros cannot be reached from this valid start; palette zeros were excluded separately. These assertions are preserved at each step of the arbitrary path.

If the original upper word is nonzero, every represented upper word is equivalent to it and cannot contain an upper zero. Hence a zero-task success cannot occur either. No lower zero declaration is reachable. Lemma 16.1 proves that the encoded word is nonzero. Whole-word endpoints do not give any additional moves: an absent cell can only prevent a proposed acquisition.

There is a nonzero letter at each level. Choose a legal idle core letter $C(r)$ with $1 < r < q$, and legal data. It is not a unary zero, and the only unary transaction creation requires $C(1)$. No binary move or zero applies to a one-letter word. It is therefore nonzero. Repeatedly encoding such letters gives nonzero words of unbounded lengths at a fixed level, using only finitely many encodings at any one time.

16. Arbitrary words and internal normalization

We must also control words with control states which never arose from clean blocks. In a nonzero word every local format and adjacency is legal, since their violations are declared zeros. Put a cut immediately before a $C(1)$ letter if its transaction track is idle or its relative block number is 1. This condition uses only that single letter.

Any palette fragment has at most one transaction head, constant-metadata wings and possibly an idle suffix; no transaction can begin inside it. It also has at most one checker head and constant passed and idle runs. Removing the at most two head cells and refining these ordered partitions gives at most 16 intervals with identical non-color fields. This loose bound covers truncated fragments too. If a color repeated in one interval, its allowed swaps would bring equal colors adjacent and make the word zero. Each interval therefore has length at most q. A palette fragment has length at most $16q + 2$, and a block or partial block at most $17q + 3$.

Between core beginnings with no intervening cut, the relative block number must strictly increase, and cannot exceed k. An idle block or a new transaction would instead create a cut. Thus an interval without a cut has length at most

$$B_{\mathrm{h}} = (k+2)(17q+3),$$

including its two possible end fragments. The unary exclusions on a first R and a last L keep this bound valid at endpoints where a neighbor is absent.

An interval between consecutive cuts starts at $C(1)$ and ends at S. If its first block is idle, no locked region can start inside it and a following block would introduce another cut. Otherwise its first cell has relative block number 1 and is L, B or F, never R. A chain of left wings must reach a head inside the interval: it cannot exit at an external interface, cross the next cut internally, or be L at its known last position. A full head forces its right wing to that last position. A cursor leaves the remainder of its current block idle; the next block would be another cut. Hence the interval contains one through k whole structural blocks and at most one transaction.

Remove each checker by its backward moves. Then roll back the transaction or release the cursor as already proved. If this exposed a zero, the original containing word would be zero by Lemma 16.1, a contradiction. Thus release succeeds. It uses only cells in the interval, as do checker removal and subsequent palette sorting. With both tracks idle, the palette test forces exactly one of each color. Sorting them gives precisely $E_{\mathrm{h}}(u)$ for the upper data word u, with $1 \le |u| \le k$. No reachable-state assumption entered this rollback argument.

16. Completion of the hierarchy proof

There are finitely many literal cut intervals, since they contain at most k bounded blocks. Let the token set consist of the nonzero intervals of the form just analyzed, and for each choose one internally obtained upper normal form u_{t}. This finite set need not be decided by the rule program: it is used only in the proof and in finite sums. A canonical block has length L_{h}; length-preserving normalization gives

$$[t] = [E_{\mathrm{h}}(u_{\mathrm{t}})], \quad |t| = L_{\mathrm{h}}|u_{\mathrm{t}}|, \quad 1 \le |u_{\mathrm{t}}| \le k.$$

A sufficiently long nonzero word has at least two cuts by the bound on intervals with no cut. Its prefix before its first cut and suffix beginning at its last cut are bounded by B_{h}; all intervening intervals are tokens. Because cuts are determined by single letters, this literal decomposition is unique. Restricting to a subword cannot create or shift a cut.

Consider a complete palette in any such token before normalization. Structural tags and lengths are unchanged by normalization, so this palette already has length q. At most two cells are heads, and at most 16 non-head intervals have homogeneous non-color fields. One of them has length at least $(q - 2)/16$, which is at least $q/100$ for the chosen parameters. Its adjacent transpositions are explicitly in the table. Every eligible run, not just this selected one, has distinct colors by the same repetition-to-zero argument. Eligibility and maximal endpoints test only the non-color fields and head flags.

This proves all three assertions of Theorem 16.2. Its word graph and every proof above are independent of the field. In particular, changing from characteristic two to any other characteristic does not change which word classes are zero or the token equalities. □

17. Cancellation over an arbitrary prime field

For a fixed prime p, work over any field K of characteristic p. Put $h_j = h_0 D^j$, $q_j = 2^{h_j}$, $L_j = 2q_j + 1$ and $A_j = A_{h_j}(K)$. Write E_j for the corresponding substitution. A letter test assigns d by d matrices $M(a)$ over K to the letters at one level, extends multiplicatively to words, and forms

$$S_{j,n} = \sum_{|w|=n} M(w) \otimes [w]_j.$$

A chunk test is the finite labeled list and length sum of Section 7, now at level j. Distinct identifiers, including repeated word labels and different segmentations of the same word, index distinct summands. Its dimension is d and its label length bound is b.

Lemma 17.1 (Direct cancellation). *If $q_j/100 > b((p-1)d^2+3)$, every such chunk test vanishes at every sufficiently large word length.*

Proof. Discard the summands whose words are zero. In each remaining word erase only the color fields and choose the first maximal eligible run of length at least $q_j/100$. Existence follows from Theorem 16.2, and this choice depends only on the color-erased word. Select all complete chunks inside this run; at most two chunks cross its ends. If the run has length r and the selected packet has m chunks, $r \le mb + 2b$, so $m > (p-1)d^2$.

Permute these complete chunks, preserving the exterior chunks. Their total interval does not change, and all its letters have the same non-color fields. Therefore the color-erased word, the selected run, and the exterior crossing chunks stay fixed. The selected packet in the new segmentation is exactly the packet being permuted. Allowed adjacent letter swaps show that all its words have the same nonzero algebra image. Thus no member is lost among the discarded summands.

The labels in a packet are distinct: each begins with a different color within the distinct-color run. Hence the identifiers are distinct and the permutation action is free, even when the original test allowed repeated labels. Applying the selection again to any member yields the same orbit. We have a genuine partition into full permutation orbits. Each orbit contributes

$$U\Big(\sum_{\sigma \in S_m} X_{\sigma(1)} \cdots X_{\sigma(m)}\Big) V \otimes [w].$$

Lemma 6.1 makes this zero in characteristic p. Summing the finitely many orbits proves the assertion. □

17. Passing between chunks and letters

Lemma 17.2 (Cut reduction). *If all dimension-d chunk tests of label bound k eventually vanish at level j+1, all dimension-d letter tests eventually vanish at level j.*

Proof. Fix a letter test M. Use literal tokens t as chunk identifiers upstairs, with label u_t fixed by Theorem 16.2 and weight $M(t)$. Let its length sums be C_m. Let P range over words of length at most B_j with no cut letter, including the empty word. Let S range over nonempty words of length at most B_j beginning with a cut letter and containing no other cut letter. For every sufficiently large n, uniqueness of the literal cut decomposition gives

$$S_{j,n} = \sum (M(P)\otimes[P]) \cdot (\mathrm{id}\otimes E_j)(C_m) \cdot (M(S)\otimes[S]),$$
$$m = (n-|P|-|S|)/L_j \in \mathbb{N}_0.$$

Every nonzero word occurs once; extra concatenations which are zero contribute zero. The homomorphism and the token equalities give the algebra factor with the displayed order of matrix weights. The upper length tends to infinity uniformly over the finite choices of ends. Eventual vanishing of C_m proves the claim. □

Lemma 17.3 (Finite automaton). *Over an alphabet of size s, a dimension-d chunk test with maximum label length b is a fixed d by d block of a letter test of dimension at most $bs^b d$, at every word length.*

Proof. First sum weights with identical labels; distributivity preserves all chunk sums. Use states consisting of the empty word and all nonempty proper prefixes of retained labels. On reading a letter a from a state v, add an identity-weighted transition to va if this is a proper prefix, and a transition weighted by the label weight back to the empty state if va is a complete label. Both transitions may coexist. Sum weights for identical edges. The block matrix of these transitions is the letter weight.

Paths from the empty state back to itself correspond exactly to segmentations into labels: the return steps close the successive chunks and multiply their weights in the original order. Therefore its empty-to-empty block is the required chunk sum after summing words. There are at most $\sum_{i=0}^{b-1} s^i \le bs^b$ states. The empty-label-list case uses zero letter matrices. Padding with zero blocks permits any larger target dimension for positive-length sums. □

17. Termination of the cancellation procedure

Theorem 17.4. *At level zero every fixed letter test, and every fixed finite chunk test, eventually vanishes over every field of characteristic p.*

Proof. Start with a letter-test dimension d at level zero. Define the dimensions of the successive upper chunk tests by

$$d_j = d \prod_{i=1}^{j-1} (k|\Sigma_i|^k) \quad (j \geq 1).$$

Cut reduction first gives dimension $d_1 = d$ upstairs. At each further step the automaton changes a bound-k chunk test at level i to a letter test of dimension at most d_{i+1} at that same level, and cut reduction moves it up one level as a chunk test. Since $|\Sigma_i| = 2^{ch_i}$,

$$\log_2(d_j^2) \leq 2 \log_2 d + 2(j-1)\log_2 k + (2kc/(D-1))(h_j - h_1).$$

The coefficient $2kc/(D-1)$ is less than one. Thus $h_j - \log_2(d_j^2)$ tends to infinity. For the fixed prime p, at some finite level J one has

$$q_J/100 > k((p-1)d_J^2+3).$$

Lemma 17.1 now kills every test of the required dimension and bound at level J. Apply Lemmas 17.2 and 17.3 backwards through the finitely many levels. They give eventual vanishing at level zero. Each test has its own finite threshold: the finite end lists and their length dilations only change that threshold finitely many times. No common bound over all dimensions or matrices has been used.

Finally any fixed chunk test at level zero is a block of a letter test of some finite dimension by Lemma 17.3. Its eventual vanishing follows from the letter-test assertion. □

This argument is the only cancellation step in which the characteristic enters. The factor $p - 1$ is fixed during the passage up the hierarchy and is absorbed by the growing ratio q_j/d_j^2. The finite presentation at level zero itself is independent of p.

18. The constructed algebra for every prime

Theorem 18.1. *For every prime p, the finite level-zero rule table of Section 16 defines an infinite-dimensional finitely presented graded unital associative algebra* $R(p) = F_p \oplus I(p)$*, with finite-dimensional homogeneous parts and matrix-nil positive ideal I(p).*

Proof. Set $R(p) = A_{h_0}(\mathrm{F_p})$. Its alphabet and complete rule list are finite, and every relator has positive homogeneous degree. The empty word spans its degree-zero copy of $\mathrm{F_p}$. There are only finitely many words of each fixed length, so every homogeneous part is finite-dimensional.

Choose a nonzero letter at level j. Iterating $E_0, \ldots, E_{j-1}$ produces a nonzero word in $R(p)$ of length $L_0 \cdots L_{j-1}$. These degrees strictly increase. By the grading their nonzero images are linearly independent, proving infinite dimension.

Let T be any finite matrix over $I(p)$. Expand its entries as a finite sum of nonempty word images with numerical matrix coefficients. By Lemma 16.1, extending from $\mathrm{F_p}$ to $\mathrm{F_p}(z)$ preserves the same word basis and all its zero classes. Apply Theorem 17.4 to the chunks in that expansion with each weight multiplied by the central indeterminate z. Proposition 7.1 then extracts the coefficient corresponding to the number of matrix factors and gives $T^a = 0$ for all sufficiently large a. This holds for every finite matrix size and every such T, proving matrix nilpotence. □

The definition of $R(p)$ uses only the finite level-zero presentation. Higher levels supply proofs of its properties; they are not an infinite family of defining relations. All roots and groups in the remaining sections use this particular algebra. Write $R = R(p)$ and $I = I(p)$ from now on. For $d \geq 1$ the notation $R_{\geq d}$ also means $I_{\geq d}$.

19. Compatible phases on long nonzero words

Lemma 19.1. *For $Q = L_0 \cdots L_d$, there is a threshold N such that each nonzero basis word of length at least N has phases $\sigma(w)$, $\tau(w)$ in $\mathbb{Z}/Q\mathbb{Z}$, constant on its word class, with $\tau(w) = \sigma(w) + |w|$. If v is a subword of such a word w, begins at offset b and has length at least N, then $\sigma(v) = \sigma(w) + b$.*

Proof. First construct phases for literal nonzero words, going downwards from level $d + 1$, where the modulus is 1 and the length threshold is 1. Suppose level $j + 1$ has phases modulo $Q' = L_{j+1} \cdots L_d$ with threshold n_{j+1} and the subword property. For the token decomposition $w = P_w t_1 \cdots t_r S_w$, use the previously fixed upper words and put $U(w) = u_{t_1} \cdots u_{t_r}$.

Choose n_j above the token threshold and at least $2B_j + L_j n_{j+1}$. Then $|U(w)| \geq n_{j+1}$. It is nonzero: otherwise the token equalities and the algebra homomorphism E_j would make w zero. Define

$$\sigma_j(w) = L_j \sigma_{j+1}(U(w)) - |P_w| \quad \text{modulo } L_j Q'.$$

The factor L_j makes this independent of the chosen representative of the upper residue. If v is a long enough subword at offset b, its cuts are exactly the original cuts lying in it. Its complete tokens are a consecutive subsequence of those of w, so $U(v)$ is a literal subword of $U(w)$. If its upper offset is a, the lengths satisfy

$$b + |P_v| = |P_w| + L_j a.$$

The upper subword identity substituted into the definition now gives $\sigma_j(v) = \sigma_j(w) + b$. This completes the downward induction for literal words.

At level zero increase the threshold to $N = 2n_0 + k$. A single elementary replacement touches at most k letters, so an unchanged prefix or suffix has length at least n_0. It has the same offset before and after the replacement. The literal subword property applied to this common piece makes the phases of the two words equal. Every word in a nonzero replacement class is nonzero, so chaining proves invariance on the entire class. Length invariance gives the same for τ. The subword identity survives the increased threshold. □

19. Prime-to-p phase moduli

For long nonzero words u,v, a nonzero product has matching middle phases and inherits the outside phases. Indeed, apply the subword identity to the prefix and suffix of their concatenation. Thus

$$\tau(u) \neq \sigma(v) \quad \Rightarrow \quad uv = 0;$$

$$uv \neq 0 \quad \Rightarrow \quad \sigma(uv) = \sigma(u), \quad \tau(uv) = \tau(v).$$

These implications also hold after reduction of the phases modulo any divisor of Q, by Lemma 8.3. In the current hierarchy $L_j = 2^{h_j+1} + 1$. Apply Lemma 8.1 with $m = h_j + 1$, and for every d put

$$Q_d = \prod_{j=0}^{d} L_j, \quad \ell_d = Q_d / p^{v_p(Q_d)}.$$

Corollary 8.2 proves that ℓ_d is coprime to p and tends to infinity. Reducing the phases supplied by Lemma 19.1 modulo ℓ_d gives compatible start and end phases modulo these arbitrarily large prime-to-p integers. The length threshold may depend on the chosen modulus. No common threshold over all moduli is asserted.

The following group argument uses one such modulus ℓ and its corresponding threshold N. It will turn the phase identities into an actual identity $z^{\ell} = 1$ in the stable Steinberg group. This is the additional group assertion needed by Proposition 9.1.

20. Long entries and opposite roots

Work over the particular algebra $R(p)$ of Theorem 18.1. Fix one phase modulus ℓ supplied by Section 19, its threshold N, and an integer $M \geq 2N + k + 1$. Every group in Sections 20-22 is stable. A nonzero word is also used to denote its basis class. Its length and its long-word phases are well defined on that class.

Lemma 20.1 (Moving a cut). *If $ca = 0$, then for three distinct indices i,h,j and arbitrary b,*

$$[x_{ij}(ab), x_{ji}(c)] = [x_{ih}(a), x_{hi}(bc)].$$

In particular $[x_{ij}(a),x_{ji}(c)]$ is independent of j. It centralizes both factors, has order dividing p, and has matrix image $1 + acE_{ii}$.

Proof. Apply the three-factor identity of Section 10 to $X = x_{ih}(a)$, $Y = x_{hj}(b)$, $Z = x_{ji}(c)$. Their consecutive commutators are $A = x_{ij}(ab)$ and $B = x_{hi}(bc)$. The required commuting pairs follow from their positions, or from $ca = 0$ and $(bc)(ab) = 0$. The identity follows. Setting $b = 1$ changes the auxiliary index arbitrarily. Use an index outside $\{i,j\}$ to express this commutator; each of its two factors commutes with both original factors, since the only possible consecutive parameter is ca. If C is the commutator, $[x_{ij}(a)^p,x_{ji}(c)] = C^p = 1$. Finally multiply the four elementary matrices; $ca = 0$ cancels all extra entries except acE_{ii}. □

Lemma 20.2. *For a nonzero class w with $|w| \geq M$ and $\sigma(w) \neq \tau(w)$, there is an element $H_i(w)$, independent of any auxiliary coordinate, such that whenever u,v are nonzero words with $|u|,|v| \geq N$, $|u|+|v| \geq M$ and $\sigma(u) \neq \tau(v)$,*

$$[x_{ij}(u),x_{ji}(v)] = H_i(uv) \quad \text{if } uv \neq 0,$$
$$[x_{ij}(u),x_{ji}(v)] = 1 \quad \text{if } uv = 0.$$

The element $H_i(w)$ has order dividing p and matrix image $1+wE_{ii}$. The zero alternative is a group assertion; vanishing matrix products alone would not justify commutation of opposite roots.

20. Independence of the word and its cut

Proof of Lemma 20.2. The mismatch implies $vu = 0$ by Section 19. To move a cut right, write the word as abc with $a = u$, $bc = v$ and $|c| \geq N$. Its end phase is $\tau(c) = \tau(v)$, so $ca = 0$. Lemma 20.1 identifies the commutators for $a \mid bc$ and $ab \mid c$. A move left uses the same formula with $ab = u$ and $c = v$. The outer phases stay unchanged when the new parts are nonzero. If either new part is zero, the equality already proves the original commutator trivial.

For a word of length $n \geq M$, legal cut positions range from N to $n - N$. A relation of length at most k straddles at most $k - 1$ of them. Since $M \geq 2N + k + 1$, choose a legal cut for which that relation lies wholly in one part, moving the cut as above. Replacing within that part leaves its ring element and, if nonzero, its phase unchanged. Thus the commutator is preserved under any replacement, unless it has already been proved trivial by a zero part.

If the concatenation is nonzero, every equivalent word and every one of its factors is nonzero. Cut movement and replacement therefore prove dependence only on the concatenation class. For any class w in the statement, choose a cut with both lengths at least N. Its parts inherit $\sigma(w)$ and $\tau(w)$, so the mismatch holds; define $H_1(w)$ by that commutator.

If the concatenation is zero, Lemma 16.1 supplies a replacement path to a word containing a declared zero factor. Follow the path with movable cuts. If neither part became zero earlier, choose the final cut with the zero factor entirely in one part. That part is zero in the algebra, so its root is the identity and the commutator is one. The order and matrix image statements follow from Lemma 20.1. □

A subgroup on which the matrix map is faithful

Let S be a finite coordinate set with assigned phases $P(i)$ modulo ℓ. A long entry w from i to j is consistent if $\sigma(w) = P(i)$ and $\tau(w) = P(j)$. Call it a wrong-row entry if instead

$$|w| \geq M, \quad \sigma(w) \neq P(i), \quad \tau(w) = P(j).$$

Use $x_{ij}(w)$ for a wrong-row entry off the diagonal and $H_1(w)$ on the diagonal. Let $U(S,P)$ be the subgroup they generate.

20. Faithfulness and normalization of the extra factors

Lemma 20.3. *The group U(S,P) is abelian of exponent dividing p, its matrix map is injective, and every consistent root normalizes it.*

Proof. For an off-diagonal wrong-row entry $x_{ij}(a)$ and another off-diagonal root $x_{ef}(b)$ with wrong-row or consistent entry, a forward consecutive commutator has parameter ab. It is zero for a wrong-row b, since $\tau(a) = P(j) \neq \sigma(b)$. For consistent b, a nonzero ab inherits the outside phases and is again wrong-row. A reverse consecutive product ba is always zero since $\tau(b) = P(i) \neq \sigma(a)$.

For opposite positions the same last mismatch permits Lemma 20.2. The commutator is $H_i(ab)$ if ab is nonzero and is one otherwise; in the former case it is a diagonal wrong-row entry. If both roots are wrong-row, both products are zero. All remaining pairs commute by their positions. This treats all off-diagonal generators.

Represent a diagonal $H_i(w)$ as a commutator through a coordinate outside S. A root on S disjoint from i commutes with it. For a root involving i, the faithful strip between this two-coordinate support and its complement gives the exact conjugation rules

$$H_i(w)\, x_{if}(b)\, H_i(w)^{-1} = x_{if}(b+wb),$$

$$H_i(w)\, x_{ei}(b)\, H_i(w)^{-1} = x_{ei}(b-bw).$$

Indeed $w^2 = 0$ by its phase mismatch, so the inverse matrix has diagonal correction $-w$. The incoming product bw is zero by phases. The outgoing product wb is zero for wrong-row b and, when nonzero for consistent b, is wrong-row. Hence these formulas prove the required commutations and normalization for diagonal generators. The minus sign is retained for odd characteristic.

Diagonal generators at different coordinates can be represented on disjoint pairs of coordinates. At the same coordinate, write a second generator as $[x_{it}(a),x_{ti}(c)]$ with $ac = w'$ and both parts of length at least N. Use a new auxiliary t. Their phases give $wa = cw = 0$, so the first generator fixes both factors by the same strip action. Thus all generators of U commute and have order dividing p.

Every pair of correction matrices multiplies to zero: a coordinate match forces the end phase of the first word to differ from the start phase of the second. A collected group element therefore has matrix $1 + \sum c_{ijw} wE_{ij}$, with coefficients in F_p. The nonzero word classes form a basis, so an identity matrix forces all coefficients zero. The collected element is then one. Finally the commutator computations give conjugation into U by consistent roots; their inverses are their $(p-1)$st powers, so they normalize U. □

21. Relocating a central element according to phases

Lemma 21.1. *Suppose $z \in J(R)$ has a root-word representative $\prod_v x_{iv,jv}(w_v)$ in nonzero word classes of lengths at least M. In coordinates (i,s), with $s \in \mathbb{Z}/\ell\mathbb{Z}$,*

$$z = \prod_v x_{(iv,\sigma(wv)),(jv,\tau(wv))}(w_v).$$

Proof. Let $A = F_p \oplus R_{\geq M}$. Choose numerical columns u_s and rows v_s of size ℓ with $v_s u_s = 1$. On its long-word basis set

$$\rho(w) = u_{\sigma(w)} v_{\tau(w)} w, \quad \rho(1) = 1_\ell.$$

This extends to a unital homomorphism $A \to M_\ell(R)$. A nonzero product has matching middle phases, so its middle numerical pairing is one and its outside phases are correct. A zero product contributes the zero entry regardless of the numerical pairing. Linear extension and scalar multiplication complete the check. The original root word has identity matrix already over A, since A embeds in R. Hence every such evaluation followed by block roots belongs to $J(R)$. No injectivity of $St(A) \to St(R)$ is needed.

Begin with all $u_s = e_0$ and all $v_s = e_0^*$. The evaluated word is the coordinate copy of z in inner index zero and hence equals z by centrality and Lemma 11.3. First change the rows to $v_0 = e_0^*$ and $v_t = e_0^* + e_t^*$ for $t \neq 0$. The added factors run from occupied inner-zero coordinates to unused coordinates. They lie in a faithful strip normalized by the original roots. Collect them on the right. Both old and new evaluations have identity matrix, so the strip correction is one. The evaluation is still z.

Let Q have these new rows. It is invertible, $v_t Q^{-1} = e_t^*$, and $Q e_0 = \sum_r e_r$. Apply the exact numerical conjugation of Section 14 to replace the columns by this sum and the rows by e_t^*. Centrality leaves the evaluation unchanged. Each block root is now $\prod_r x_{(i,r),(j,\tau(w))}(w)$. Give coordinate (i,s) phase s. Keep its consistent factor with $r = \sigma(w)$ and collect all other factors into U by Lemma 20.3. The retained word has identity matrix too, by the homomorphism ρ with columns e_s and rows e_s^*. The collected U-element has identity matrix and is one. This proves the formula. □

21. Cyclic evaluation is a power in the same group

Proposition 21.2. *For z with a representative as in Lemma 21.1, its graded cyclic evaluation satisfies $W_z(C_\ell) = z^\ell$ in St(R).*

Proof. A homogeneous word w has degree $|w|$. Its graded evaluation at C_ℓ gives a block root with entry $wC_\ell^{|w|}$. Since $\tau(w) - \sigma(w) = |w|$ modulo ℓ, its constituent factors can be written

$$x_{(i,\sigma(w)+a),(j,\tau(w)+a)}(w), \quad a \in \mathbb{Z}/\ell\mathbb{Z}.$$

Factors with different offsets commute, including factors coming from different original roots. If the column of one matches the row of another, $\tau(w) + a = \sigma(w') + a'$. For different offsets this means $\tau(w) \neq \sigma(w')$, so $ww' = 0$. A reverse match similarly implies $w'w = 0$. The defining root relations prove commutation unless the positions are opposite. In that case the reverse match also gives $\sigma(w) \neq \tau(w')$; Lemma 20.2, with its zero-product alternative, proves the needed group commutation.

Collect by offset, retaining the original order inside each offset. Offset zero is the phase-relocated representative and hence equals z by Lemma 21.1. Every other offset is its coordinate permutation and also equals z by Lemma 11.3. There are ℓ offsets, proving $W_z(C_\ell) = z^\ell$. □

All statements so far were for roots whose parameters are basis word classes. General coefficients in F_p cause no restriction: write a scalar coefficient as an integer between 0 and $p - 1$ and use root addition to replace it by that many occurrences of the root. Inverse roots use $p - 1$ copies. This preserves the degree lower bound and gives a finite word, so the arguments apply in every prime characteristic.

22. Long representatives and stable separation

Lemma 22.1. *Every element killed by St(R) → St(R/$R_{\geq 2M}$) admits a finite representative in roots of nonzero word classes of lengths at least M.*

Proof. For any ideal K the kernel of $\mathrm{St}(R) \to \mathrm{St}(R/K)$ is the normal closure of its roots with parameters in K. Killing these roots makes parameters depend only on their residue class by addition, and the remaining presentation is exactly that of the quotient ring. For $K = R_{\geq 2M}$, expand every parameter in the word basis and its coefficients as repeated roots. It suffices to consider $g\mathrm{x}_{ij}(w)g^{-1}$ with $|w| \geq 2M$.

Split a representative $w = ab$ with both parts of length at least M. They are nonzero because w is nonzero. Let S contain i,j and the finite support of a fixed root word for g, and choose h outside S. Then

$$g\mathrm{x}_{ij}(w)g^{-1} = [g\mathrm{x}_{ih}(a)g^{-1}, g\mathrm{x}_{hj}(b)g^{-1}].$$

Writing A_g for its matrix on S, exact strip conjugation gives respective root parameters $(A_g)_{ri}a$ and $b(A_g^{-1})_{js}$. They all belong to the ideal $R_{\geq M}$. Expand the strips, their entries and the commutator as a finite root word; inverse coefficients can again be expanded into repeated roots. Every entry has length at least M. Apply this to the finitely many conjugates in a normal-closure expression. □

Theorem 22.2 (Stable separation). *For every prime p, the constructed algebra R(p) satisfies*

$$\cap_{d\geq 1} \ker[\mathrm{St}(R(p)) \to \mathrm{St}(R(p)/R(p)_{\geq d})] = \{1\}.$$

Proof. Let z belong to the intersection. Its finite matrix image is the identity modulo every degree ideal, whose intersection is zero. Thus $z \in J(R)$; truncation at degree one gives trivial reduction in $\mathrm{St}(\mathrm{F}_p)$. Theorem 14.1 gives $z^{p^a} = 1$. Fix one root word for z. The wrap-removal argument of Section 14 supplies a bound beyond which $W_z(C_b) = 1$ for every integer b.

Choose a phase modulus ℓ coprime to p above that bound, using Section 19. Only now choose its phase threshold N and $M \geq 2N + k + 1$. Since z is killed at degree $2M$, Lemma 22.1 gives a long-entry representative. Proposition 21.2 proves $z^{\ell} = W_z(C_{\ell}) = 1$. The operation W_z depends on the group element, not its representative, so the earlier bound remains valid. Coprimality and Bézout's identity yield $z = 1$. □

23. Conclusion

For each prime p, the finite word-rule construction defines an infinite-dimensional finitely presented graded algebra $R(p)$. Reversible verification makes the hierarchy faithful on nonzero words, while cut normalization and palette symmetrization prove that every finite matrix over its positive ideal is nilpotent. Only the finite presentation at the first level defines the algebra; the higher levels are used to prove its properties.

The same hierarchy gives compatible phases on long word classes with arbitrarily large periods coprime to p. The long-entry and relocation arguments turn those phases into the identity $W_z(C_\ell) = z^\ell$ inside the stable Steinberg group. Wrap removal gives an annihilating such ℓ, and relative p-power torsion forces every element in all truncation kernels to be trivial. Thus the separation assertion holds for the particular algebra already constructed.

Finite presentation of Steinberg groups, fixed-rank injectivity, finiteness of their truncations, and finite-index rewriting now yield the group

$$G_p = \ker[\mathrm{St}_{12}(R(p)) \to \mathrm{St}_{12}(\mathrm{F}_p)].$$

It is infinite, finitely presented as an abstract group, residually finite, and every element has finite p-power order. These properties belong to this one group for each prime; neither a conjectural algebra input nor an additional separation hypothesis remains in Theorem 1.1. There is no claim of a common exponent for all elements.